Neutrino-Nucleon Structure Functions in Duality and DIS Regions

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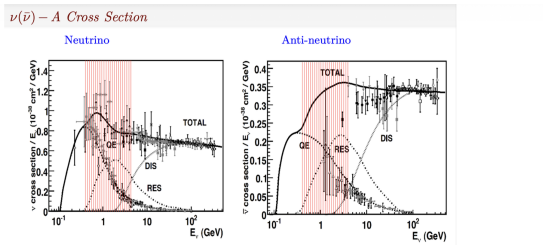
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Tsung-Dao Lee Institute, Shanghai Jiao Tong University

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Neutrino Interactions in few GeV energy region

- Neutrino interactions span:
  - Quasi-Elastic (QE)
  - Resonance (RES)
  - Deep Inelastic Scattering (DIS)
- Significant event rates occur in the few-GeV transition region which spans from resonance to DIS..
- DIS modelling impacts:
  - generators,
  - energy reconstruction,
  - oscillation analyses.



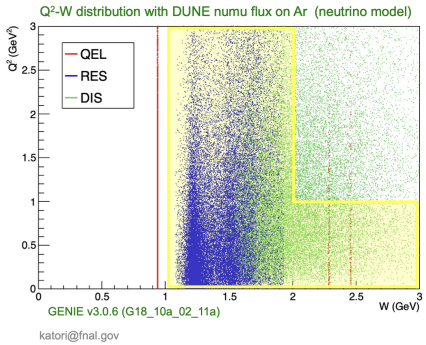
$$\sigma^{Total} = \sigma^{QE}_{\nu_l n \to l^- p} + \sigma^{Inelastic} + \sigma^{DIS} \quad \nu_l(\bar{\nu}_l)(k) + N(p) \to l^{\mp}(k') + X(p'),\ l = e, \mu$$

$$\sigma^{Inelastic} = \sigma^{1\pi} + \sigma^{2\pi} + \cdots + \sigma^{YK} + \sigma^{1K} + \sigma^{1Y} + \sigma^{\eta} + \cdots$$

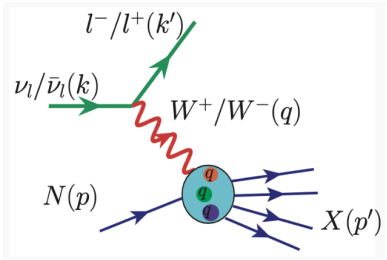
Why Low-Q^2 DIS Modelling Matters

- No single model describes the full kinematic range.
- QE, resonance, and DIS processes overlap in this region.
- Accurate modelling remains challenging at low-Q^2.

A smooth transition between low-Q^2 and perturbative DIS descriptions is needed.



Neutrino Deep Inelastic Scattering (DIS)



- In DIS, the neutrino probes the quark structure of the nucleon through interactions with large momentum transfer.
- The charged-current DIS cross section depends on the structure functions $F_1(x, Q^2)$, $F_2(x, Q^2)$ and $F_3(x, Q^2)$:

$$\frac{d^2\sigma^{\nu,\bar{\nu}}}{dx\,dy} \propto xy^2F_1 + (1-y)F_2 \pm xy\left(1-\frac{y}{2}\right)F_3$$

where,

- x : Bjorken scaling variable
- Q^2 : momentum transfer squared
- y : inelasticity variable

- F_1 and F_2 encode the vector structure of the nucleon, while F_3 arises from weak-interaction parity violation and is sensitive to valence quarks.
- Accurate modelling of low-Q^2 structure functions is important for reliable neutrino DIS cross-section calculations.

The structure functions for (anti)neutrino scattering on isoscalar targets are given by

$$F_2^{\nu N}(x,Q^2) = x(u(x,Q^2) + d(x,Q^2) + \bar{u}(x,Q^2) + \bar{d}(x,Q^2) + 2s(x,Q^2) + 2\bar{c}(x,Q^2))$$

$$F_2^{\bar{\nu} N}(x,Q^2) = x(u(x,Q^2) + d(x,Q^2) + \bar{u}(x,Q^2) + \bar{d}(x,Q^2) + 2\bar{s}(x,Q^2) + 2c(x,Q^2))$$

and

$$xF_3^{\nu N}(x,Q^2) = x(u(x,Q^2) + d(x,Q^2) - \bar{u}(x,Q^2) - \bar{d}(x,Q^2) + 2s(x,Q^2) - 2\bar{c}(x,Q^2))$$

$$xF_3^{\bar{\nu} N}(x,Q^2) = x(u(x,Q^2) + d(x,Q^2) - \bar{u}(x,Q^2) - \bar{d}(x,Q^2) - 2\bar{s}(x,Q^2) + 2c(x,Q^2))$$

Perturbative and Non-Perturbative Corrections at Nucleon Level:

Perturbative QCD corrections: Next-to-leading order corrections (NLO)

- The partons inside the nucleon may interact among themselves via gluon exchange which is described by the QCD.
- The contribution from gluon emission taken into account, then the nucleon structure function shows dependence on Q, i.e., the Bjorken scaling is violated.
- For a finite value of Q^2, strong coupling constant is large and next-to-leading order terms give a significant contribution followed by next-to-next-to-leading order terms.

The expressions for the structure functions $F_{2,L}$ and F_3, taken from the work of Vermaseren et al. and Moch et al., can be expressed as functions of the PDFs as follows:

$$x^{-1}F_{2,L} = \sum_{f=q,g}^{\infty} C_{2,L} \otimes f_2, \quad F_3 = \sum_{f=q,g}^{\infty} C_3 \otimes f_3$$

where C_2 and C_3 are the coefficient functions for the quarks and gluons, $\otimes$ denotes the Mellin convolution (which simplifies to multiplication in N-space), and f_2 and f_3 represent the quark and gluon distributions.
Vermaseren and van Neerven et al. NPB 724 (2005) 3,
Van Neerven and Vogt NPB 568 (2000) 263.

Non-Perturbative QCD corrections: Target Mass corrections (TMC)

- The TMC contributions are associated with the finite mass of the target nucleon M_N, which are relevant at high x and moderate Q^2.
- This results in the modification of the kinematics for the scattering process.
- These effects are important to determine the valence quark distribution at high x and therefore x is replaced by the Nachtmann scaling variable ξ:

$$\xi = \frac{2x}{1+\gamma}$$

For convenience, the variable γ was introduced as

$$\gamma = \sqrt{1 + 4 \cdot x^2 \cdot \mu}$$

where,

$$\mu = \frac{M^2}{Q^2}$$

$Q^2 = -q^2$, q is the momentum transfer of the leptons

The structure functions with TMC are given as

$$F_1^{\mathrm{TMC}}(x,Q^2) \simeq \frac{x}{\xi\gamma}F_1(\xi)\left[1+\frac{2\mu x\xi}{\gamma}(1-\xi)^2\right]$$

$$F_2^{\mathrm{TMC}}(x,Q^2) \simeq \frac{x^2}{\xi^2\gamma^3}F_2(\xi)\left[1+\frac{6\mu x\xi}{\gamma}(1-\xi)^2\right]$$

$$F_3^{\mathrm{TMC}}(x,Q^2) \simeq \frac{x}{\xi\gamma^2}F_3(\xi)\left[1-\frac{\mu x\xi}{\gamma}(1-\xi)\ln\xi\right]$$

Schienbein et al. "Target mass corrections." Journal of Physics G: Nuclear and Particle Physics 35.5 (2008): 053101.

Dynamical Higher Twist Effect (Significant at High x and Low Q^2)

Higher twist corrections include multiparton correlation in the nucleon

1. At low Q^2 Lepton/Neutrino-nucleon scattering can involve multiple partons.
2. Compared to the leading twist diagrams (pQCD), the higher twist diagrams are suppressed by powers of $\frac{1}{Q^2}$, so they are important at low Q^2.
3. This is also the region where the strong coupling constant is large and pQCD is invalid.

$$F_2(x, Q^2) = F_2^{LT+TMC}(x, Q^2)\left(1 + \frac{D_2(x, Q^2)}{Q^2}\right)$$

where the first term is known as the leading twist (LT) term and is responsible for the evolution of structure functions via perturbative QCD corrections. The D_2 reflects the strength of multiparton correlations (qq and qg).For Details: E. Stein et al., Nucl. Phys. B 536, 318 (1998) M. Dasgupta et al., Phys. Lett. B 382, 273–281, 1996.

Problem in the Resonance Region

- In the resonance region ($W \lesssim 2$ GeV), structure functions exhibit:
  - Pronounced peaks associated with baryon resonances (Δ, N^*, etc.)
  - Strong Q^2 dependence governed by transition form factors
- Conventional expectation:
  - Resonance contributions should be rapidly suppressed as Q^2 increases
  - A smooth deep-inelastic background should dominate
- **Bloom–Gilman observation:** the prominent resonances do *not* disappear relative to the DIS scaling curve

Main Observation: Bloom–Gilman Duality

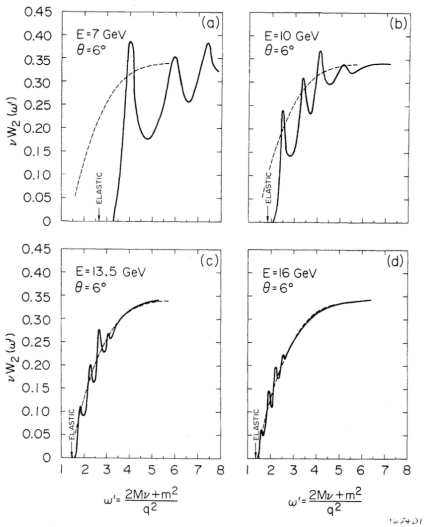
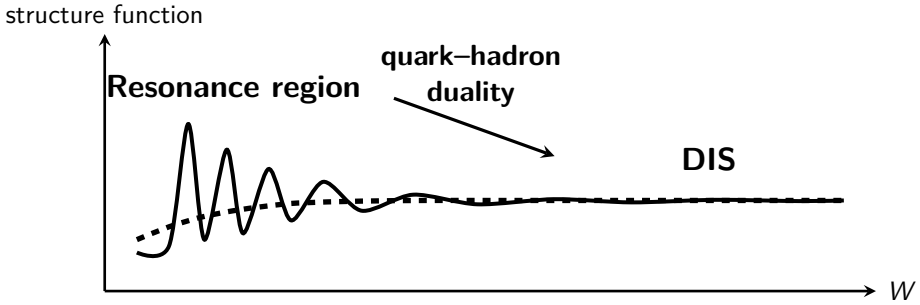


Figure 1 The function νW_2 plotted versus $\omega' = (2M\nu + m^2)/q^2$, with $m^2 = M_N^2$. The solid lines are smooth curves drawn through the $\theta = 6^0$ data at

- When plotted versus ω':
  - Resonance peaks persist with increasing Q^2
  - Resonances oscillate around the DIS scaling curve
- Each resonance follows the same envelope as DIS
- Averaged resonance strength reproduces DIS structure functions
- This phenomenon is known as **Bloom–Gilman (quark–hadron) duality**
- Resonance and DIS regimes are not independent, but reflect the same underlying dynamics

Quark–Hadron Duality

The resonance region exhibits hadronic structure, while DIS is described in terms of partons.



Duality connects the averaged resonance-region response to the DIS description.

Motivation: an alternative low-Q^2 model for generators

- Neutrino DIS covers both non-perturbative and perturbative Q^2 regions.
- This motivates matching a low-Q^2 parametrization to perturbative DIS calculations.
- We study CKMT at low Q^2 matched to AMU at high Q^2 as an alternative to Bodek–Yang.

Takeaway: build a practical low-Q^2 alternative model.

CKMT for low Q^2, AMU for high Q^2

- CKMT is used in the non-perturbative low-Q^2 region.
- AMU describes the perturbative high-Q^2 DIS region using PDFs.
- Neither model alone covers the full kinematic range needed for neutrino generators.
- Central question is how to match CKMT to AMU across the transition region.

Takeaway: the problem is not choosing one model, but matching the low-Q^2 and high-Q^2 descriptions.

CKMT: an empirical low-Q^2 parametrization

- CKMT is an empirical parametrization fitted to structure-function data.
- The curves show agreement with experimental $F_2(x, Q^2)$ data.

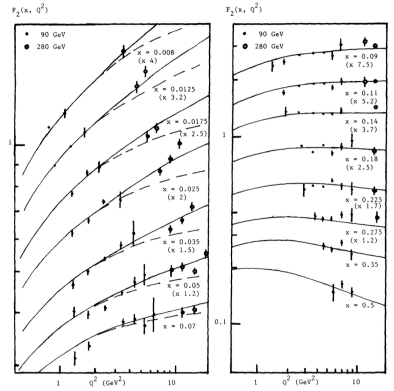


Fig. 2. The proton structure function $F_2(x, Q^2)$ obtained by the NMC collaboration [9] at 90 GeV (full circles) and 280 GeV (open circles). The dashed line is obtained from Eqs. (3) and (6) with the following values of the parameters (all dimensional parameters are in GeV^2): $A = 0.1502$, $a = 0.2631$, $\Delta_0 = 0.07684$, $d = 1.1170$, $b = 0.6452$, $\alpha_R = 0.4150$, $c = 3.5489$. (The values of the parameters B_q and B_g in (6), determined from the normalization condition for valence quarks (at $Q^2 = 2$ GeV^2), are: $B_q = 1.2064$ and $B_g = 0.1798$. The full line is obtained using QCD evolution (see Eq. (8)).

A. Capella et al., Phys. Lett. B **337**, 358 (1994)

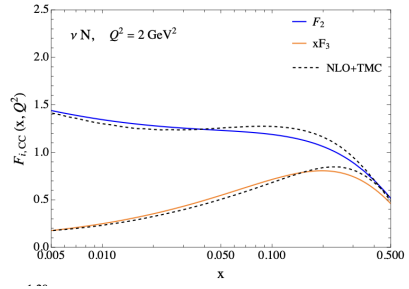
F_2 CKMT-NT (or Scaled CKMT) Parametrization

- To avoid discontinuities in the structure functions at the matching scale Q_0^2, we normalize the CKMT structure functions according to

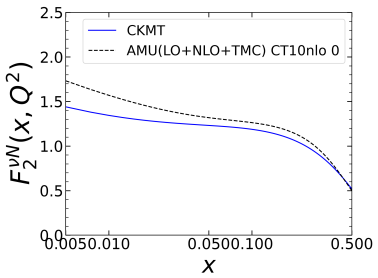
$$F_{i,\mathrm{CC}}(x,Q^2) = F_{i,\mathrm{CC}}^{\mathrm{CKMT}}(x,Q^2)\frac{F_{i,\mathrm{CC}}^{\mathrm{NLO+TMC}}(x,Q_0^2)}{F_{i,\mathrm{CC}}^{\mathrm{CKMT}}(x,Q_0^2)}, \quad (13)$$

- Results are weakly sensitive to the choice of Q_0^2 in the range 2–4 GeV^2, since the CKMT parametrization already follows the Q^2 evolution of $F_{2,\mathrm{CC}}$ reasonably well.

CKMT is validated before matching



Comparison of the NLO+TMC (black-dashed) and CKMT (blue-solid) evaluations of $F_{2,\mathrm{CC}}$

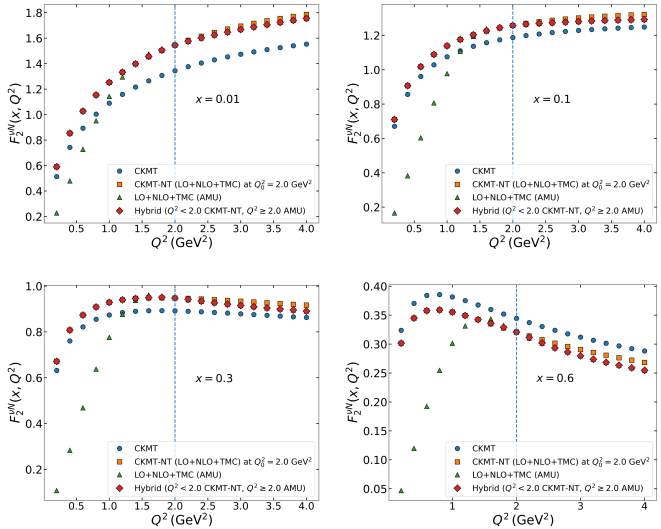


Reproduced result of CKMT F_2 (blue-solid)compared to AMU (black-dashed)with CT10nlo PDFs.

- Good agreement with published results for CKMT, Confirms correct implementation.
- Small differences in AMU come from PDF and TMC choices.

Jeong & Reno, *Phys. Rev. D* **108**, 113010 (2023)
DOI: 10.1103/PhysRevD.108.113010

Scaled CKMT matches smoothly to AMU at $Q_0^2 = 2.0\ \mathrm{GeV}^2$



$Q_0^2 = 2.0\ \mathrm{GeV}^2$ is the best satisfactory changeover point from CKMT to AMU

- Scaling CKMT at $Q_0^2 = 2.0\ \mathrm{GeV}^2$ gives a smooth transition to AMU.
- Choosing a lower transition point is unsatisfactory.
- Below this scale, AMU begins to diverge as it leaves the perturbative region.

Takeaway: we use $Q_0^2 = 2.0\ \mathrm{GeV}^2$ as the best changeover point.

Splice option: preserving low-Q^2 CKMT

Scaling CKMT at $Q_0^2 = 2.0\ \mathrm{GeV}^2$ gives a smooth transition, but it also distorts the original low-Q^2 CKMT behaviour. We therefore test an alternative in which CKMT is left unscaled and connected to AMU across a finite Q^2 window.

Linear splice prescription

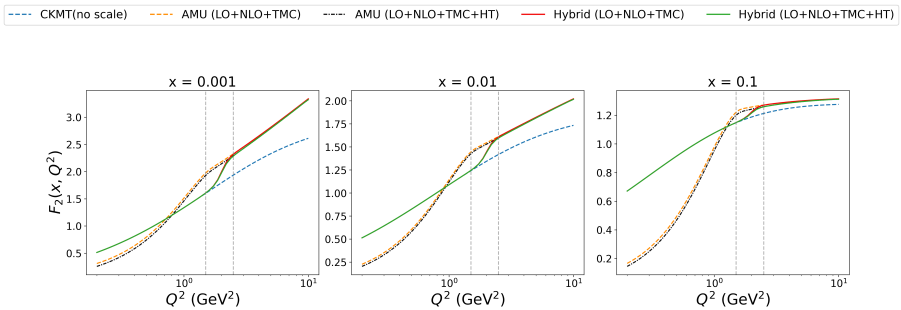
$$F_2(x, Q^2) = \begin{cases} F_2^{\mathrm{CKMT}}(x, Q^2), & Q^2 \leq 1.5\ \mathrm{GeV}^2, \\ F_2^{\mathrm{splice}}(x, Q^2), & 1.5 < Q^2 < 2.5\ \mathrm{GeV}^2, \\ F_2^{\mathrm{AMU}}(x, Q^2), & Q^2 \geq 2.5\ \mathrm{GeV}^2, \end{cases}$$

$$F_2^{\mathrm{splice}}(x, Q^2) = c(Q^2)\, F_2^{\mathrm{AMU}}(x, Q^2) + \big(1 - c(Q^2)\big)\, F_2^{\mathrm{CKMT}}(x, Q^2)$$

$$c(Q^2) = \frac{Q^2 - 1.5}{2.5 - 1.5}.$$

Takeaway: the splice favors low-Q^2 fidelity over exact smoothness.

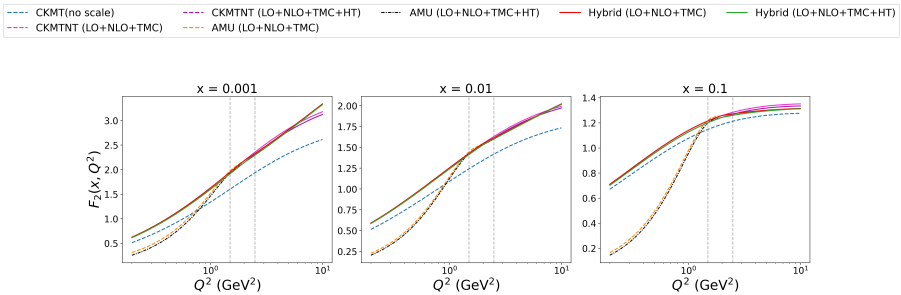
An unscaled splice improves low-Q^2 behaviour, but smoothness is lost



- The hybrid curve follows CKMT at low Q^2, then transitions to AMU across $1.5 < Q^2 < 2.5\ \mathrm{GeV}^2$.
- This keeps the low-Q^2 CKMT behavior closer to the original model.

Takeaway: the splice scheme trades smoothness for broader low-Q^2 fidelity.

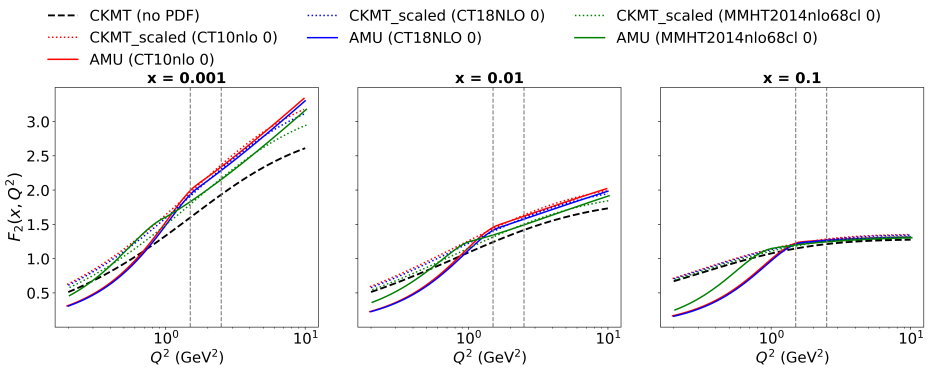
Maintaining smoothness comes at the cost of accuracy



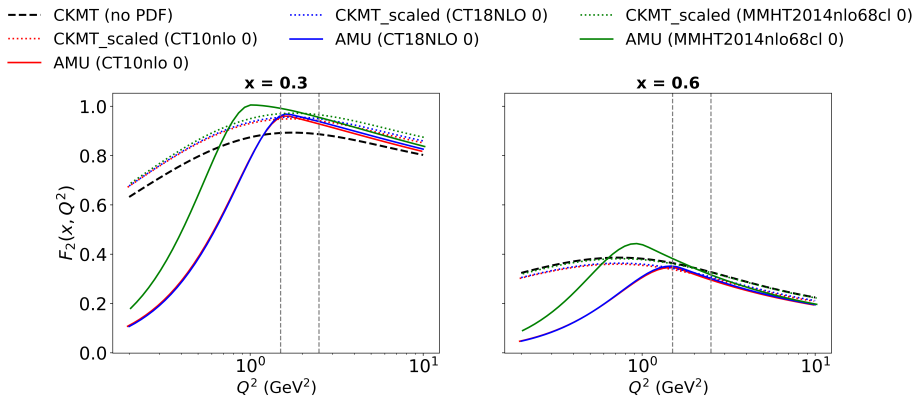
- The scaled CKMT→AMU construction is smooth
- The main choice is between smoother evolution and a wider low-Q^2 data-oriented description.

Takeaway: The apparent tension is not whether CKMT or AMU is "right," but what property we want the transition to preserve.

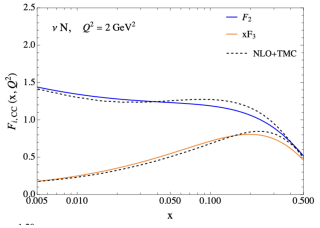
Different PDFs show significant effects at high x and low Q^2



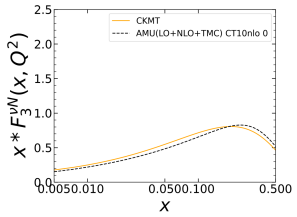
Different PDFs show significant effects at high x and low Q^2



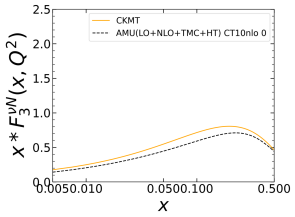
Validation of CKMT for xF_3 using Jeong and Reno Paper



CKMT vs NLO+TMC
$xF_{3,CC}$ (paper)

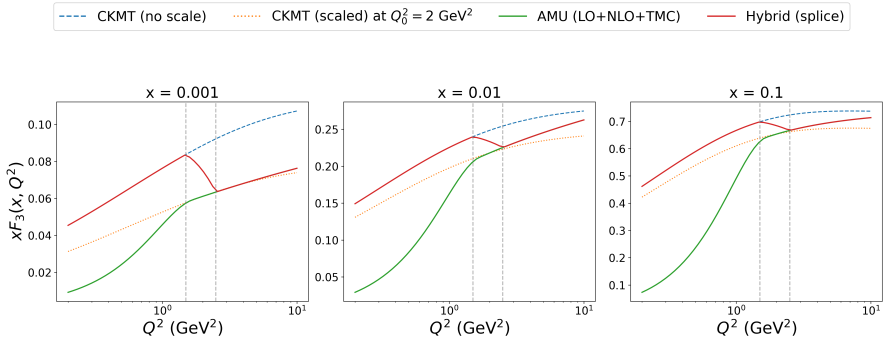


CKMT vs AMU
(LO+NLO+TMC, CT10nlo)



CKMT vs AMU
(LO+NLO+TMC+HT, CT10nlo)

Linear Interpolation Matching for xF_3 from CKMT to AMU(LO+NLO+TMC)

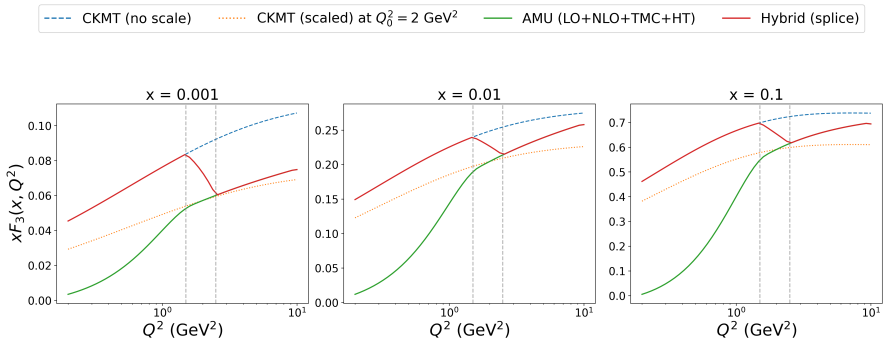


$$F_3^{\text{hyb}} = c\, F_3^{\text{AMU}} + (1 - c)\, F_3^{\text{CKMT}}$$

$$c = \frac{Q^2 - 1.5}{2.5 - 1.5}$$

- Visible kink/dip appears near the transition region.
- The effect of the matching procedure is most visible at low-x.

Linear Interpolation Matching for xF_3 from CKMT to AMU(LO+NLO+TMC+HT)



Smooth Sigmoid Matching

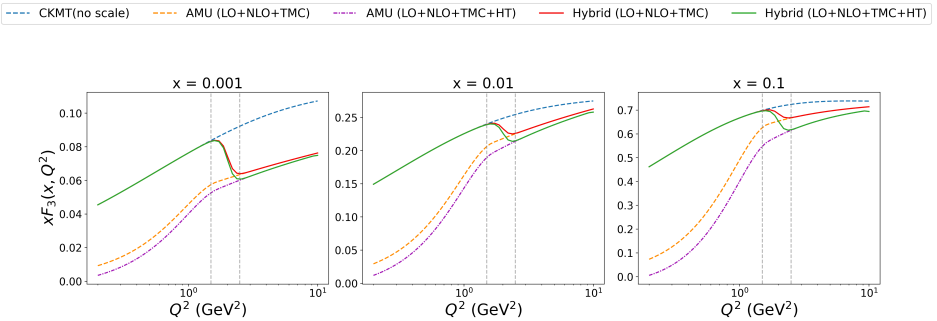
$$F_3^{\mathrm{hyb}} = w(Q^2)F_3^{\mathrm{CKMT}} + (1 - w(Q^2))F_3^{\mathrm{AMU}}$$

$$w(Q^2) = \frac{1}{2}\left[1 - \tanh\left(\frac{Q^2 - Q_0^2}{\Delta}\right)\right]$$

$$\Delta = 0.25 \quad \textit{and} \quad 0.4$$

- Provides a smooth crossover between the two regimes.
- Reduces artificial interpolation artifacts.
- Transition becomes significantly smoother.

Comparison of Hybrid Matching of xF_3: TMC vs HT



- Comparison of smooth sigmoid hybrid matching using:
  - AMU(LO+NLO+TMC),
  - AMU(LO+NLO+TMC+HT).
- HT corrections suppress the AMU prediction relative to the TMC.

Summary

- Constructed a hybrid framework for F_2, xF_3 and $2xF_1$ using:
  - CKMT at low-Q^2,
  - AMU (LO+NLO+TMC+HT) at higher Q^2.
- Linear interpolation produces visible kink structures.
- Smooth sigmoid weighting significantly improves the transition behavior.
- The transition feature remains most visible at low-x.
- The resulting hybrid structure functions can be used for neutrino DIS cross-section calculations.
- Finally, as an ongoing part of this work, we are incorporating nuclear-medium effects into the framework. The aim is to calculate the corresponding nuclear structure functions and study their impact on neutrino DIS cross sections.

Thank You

Questions?

Backup slides: CKMT Parameterization

The CKMT parametrization, obtained by fitting to electromagnetic structure function data, has the form

$$\begin{aligned} F_{2,\mathrm{EM}}^{\mathrm{CKMT}}(x,Q^2) &= F_2^{sea}(x,Q^2) + F_2^{val}(x,Q^2) \\ &= Ax^{-\Delta(Q^2)}(1-x)^{n(Q^2)+4} \\ &\times \left(\frac{Q^2}{Q^2+a}\right)^{1+\Delta(Q^2)} \\ &+ Bx^{1-\alpha_R}(1-x)^{n(Q^2)}\left(\frac{Q^2}{Q^2+b}\right)^{\alpha_R} \\ &\times \Big(1+f(1-x)\Big)\,. \end{aligned} \tag{6}$$

The functions $n(Q^2)$ and $\Delta(Q^2)$ are

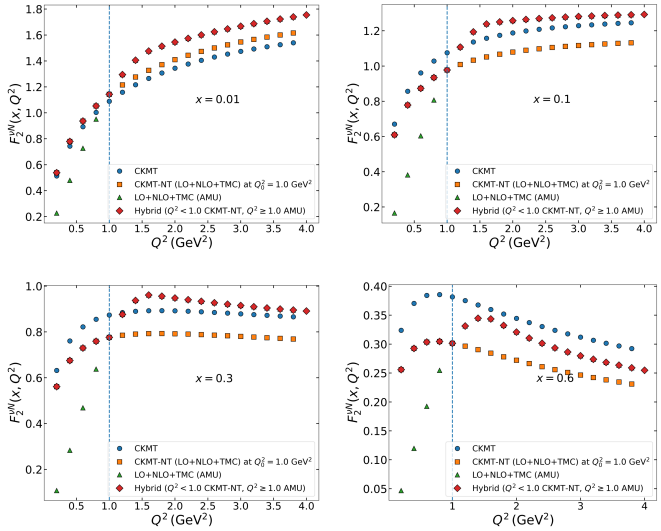
$$n(Q^2) = \frac{3}{2}\left(1+\frac{Q^2}{Q^2+c}\right), \tag{7}$$

$$\Delta(Q^2) = \Delta_0\left(1+\frac{2Q^2}{Q^2+d}\right). \tag{8}$$

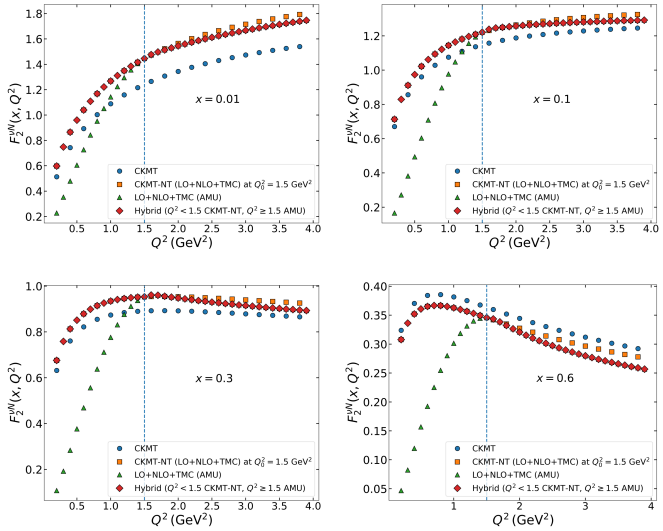
Δ_0	α_R	a [GeV^2]	b [GeV^2]	c [GeV^2]	d [GeV^2]
0.07684	0.4150	0.2631	0.6452	3.5489	1.1170
	Process	A	B	f	
	EM F_2	0.1502	1.2064	0.15	
	νN F_2	0.5967	2.7145	0.5962	
	νN xF_3	9.3955×10^{-3}	2.4677	0.5962	
	$\bar{\nu} N$ xF_3	9.3955×10^{-3}	-2.4677	0.5962	

TABLE I. Parameter values in Ref. [43] for CKMT parameterization of the electromagnetic structure function F_2. The quantities B and f are determined from the valence conditions at $Q^2 = 2$ GeV^2.

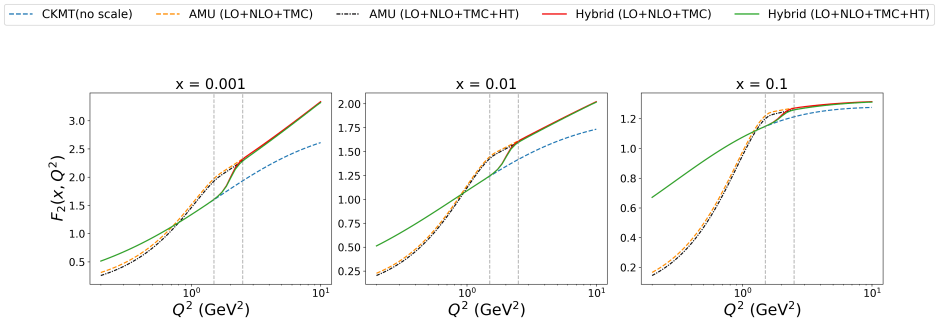
Matching at $Q_0^2 = 1.0\ \mathrm{GeV}^2$ is unsatisfactory

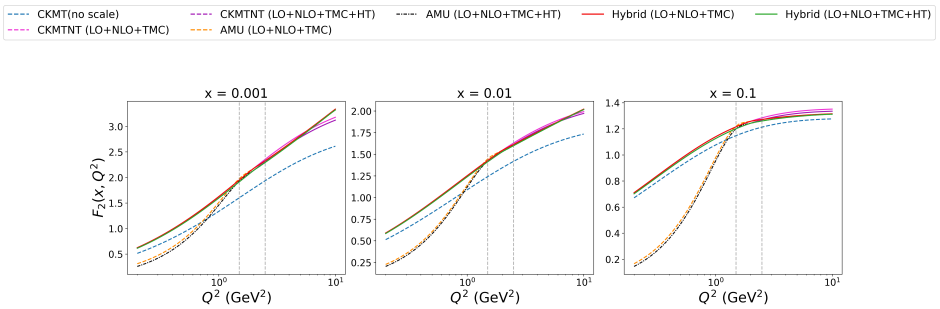


Matching at $Q_0^2 = 1.5\ \mathrm{GeV}^2$ is better, but still not ideal

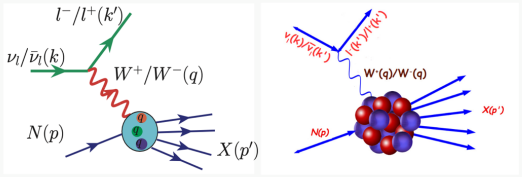


Smooth Sigmoid Matching for F_2 at $\Delta = 0.25$ with TMC and HT




CKMT(no scale)
CKMTNT (LO+NLO+TMC)
CKMTNT (LO+NLO+TMC+HT)
AMU (LO+NLO+TMC)
AMU (LO+NLO+TMC+HT)
Hybrid (LO+NLO+TMC)
Hybrid (LO+NLO+TMC+HT)
x = 0.001
x = 0.01
x = 0.1
$F_2(x, Q^2)$
Q^2 (GeV^2)

$\nu_l/\bar{\nu}_l - A$ scattering



- Fermi motion, binding energy and nucleon correlations through spectral function and is calculated using Lehmann's representation for the relativistic nucleon propagator.
- Nuclear many body theory is used to calculate it for an interacting Fermi sea in nuclear matter. A local density approximation is then applied to translate these results to finite nuclei.
- There are virtual mesons associated with the nucleon bound inside the nucleus. These meson clouds get strengthened by the strong attractive nature of nucleon-nucleon interactions.
- This leads to an increase in the interaction probability of virtual mediating quanta with the meson cloud. The effect of meson cloud is more pronounced in heavier nuclear targets and dominate in the intermediate region of $x(0.2<x<0.6)$.
- The shadowing suppression at small x occurs due to coherent multiple scattering of quark-anti quark pair coming from the virtual boson with destructive interference of the amplitudes and is incorporated following the works of Kulagin and Petti. Phys. Rev. D **76**, 094033(2007).

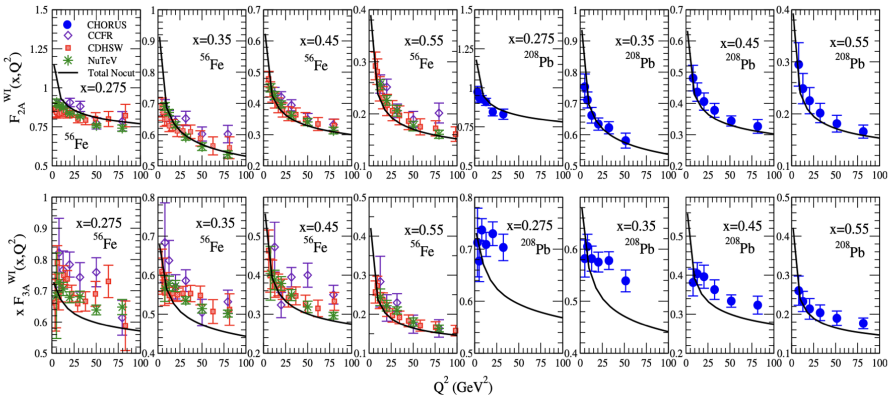
The spectral function $F_{iA,N}(x_A, Q^2)$ $(i = 1-5))$ are obtained as:

$$F_{iA,N}(x_A, Q^2) = 4 \int d^3r \int \frac{d^3p}{(2\pi)^3} \frac{M_N}{E_N(\mathbf{p})} \int_{-\infty}^{\mu} dp^0 \, S_h(p^0, \mathbf{p}, \rho(r)) \times f_{iN}(x, Q^2),$$

In our model, the nuclear structure functions is given by:

$$F_{iA}(x_A, Q^2) = \underbrace{F_{iA,N}(x_A, Q^2)}_{\text{spectral function}} + \underbrace{F_{iA,\pi}(x_A, Q^2) + F_{iA,\rho}(x_A, Q^2)}_{\text{mesonic contribution}} + \underbrace{F_{iA,shad}(x_A, Q^2)}_{\text{shadowing}},$$

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CHORUS
CCFR
CDHSW
NuTeV
Total Nocut
$F_{2A}^{WI}(x,Q^2)$
$xF_{3A}^{WI}(x,Q^2)$
x=0.275
x=0.35
x=0.45
x=0.55
^{56}Fe
^{208}Pb
Q^2 (GeV^2)

Comparison of Nuclear F_2^{SF} with Free-Nucleon Results

